

Airframe PID Parameter Comparison

All 12 airframes from `originaldefaults.h` converted to unified float .af format. Values are in FC-native units (radians, volts, fractions — not degrees, volts×10, or percent).

Rate PID Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
RollRateKp	0.16	0.095	0.1	0.16	0.095	0.16	0.04
RollRateKd	0.006	0.004	0.0045	0.006	0.004	0.006	0

Angle PID Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
RollAngleQKp	7	5.75	7	7	5.75	7	2
RollAngleQKi	0.1	0.15	0.25	0.2	0.15	0.1	0.1
RollAngleQIntLimit	0.005236	0.002618	0.002618	0.00733	0.002618	0.005236	0.002618

Pitch Axis

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
PitchRateKp	0.16	0.095	0.1	0.16	0.095	0.16	0.04
PitchRateKd	0.006	0.004	0.0045	0.006	0.004	0.006	0
PitchAngleQKp	7	5.75	7	7	5.75	7	2
PitchAngleQKi	0.1	0.15	0.25	0.1	0.15	0.1	0.1
PitchAngleQIntLimit	0.005236	0.002618	0.002618	0.00733	0.002618	0.005236	0.002618

Yaw Axis

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
YawRateKp	0.2	0.125	0.1	0.2	0.125	0.2	0.05
YawRateKd	0.000125	0.001125	0.001125	0.001125	0.001125	0.000125	0.001125
YawAngleQKp	2.25	3	3	3	3	2.25	3
YawAngleQKi	0.25	0.25	0.25	0.25	0.25	0.25	0.25
YawAngleQIntLimit	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727	0.008727

Navigation Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
NavPosKp	0.33	0.2475	0.2475	0.2475	0.2475	0.33	0.33
NavPosKi	0.008	0.012	0.012	0.012	0.012	0.008	0.008
NavVelKp	0.6	0.6	0.3	0.6	0.6	0.6	0.6

Altitude Gains

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
AltPosKp	0.5124	0.5124	0.5124	0.1464	0.4575	0.915	0.2928
AltPosKi	0.0046	0.0046	0.0046	0.00092	0.00184	0.0138	0.00552

Cruise & Limits

Param	150mm Brushed	DevEBox	Ecks 220mm	Ken's 450-1165	Ken's Alpha	Ken's LadyBug	Phoenix FW
EstCruiseThr	0.59	0.3	0.54	0.31	0.3	0.54	0.2
MaxRollAngle	1.0472	1.0472	0.523599	1.0472	1.0472	1.0472	1.0472
MaxPitchAngle	1.0472	1.0472	0.523599	1.0472	1.0472	1.0472	1.0472
NavMaxAngle	0.261799	0.349066	0.436332	0.349066	0.349066	0.261799	0.349066
BatteryCapacity	22	22	22	22	22	22	22
LowVoltThres	3.4	9.6	9.6	13.6	13.8	3	10.2
RCChannels	7	7	7	7	7	7	7
MaxRollRate	10.472	10.472	10.472	10.472	10.472	10.472	10.472
MaxPitchRate	10.472	10.472	10.472	10.472	10.472	10.472	10.472